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# Title: To create a 4 bit ripple carry adder in Logisim using only Basic Gates

# Theory:

Full Adder is a combinational circuit that adds three inputs and produces two outputs. The first two inputs are A and B and the third input is an input carry as C-IN. The output carry is designated as C-OUT and the normal output is designated as S which is SUM.

- The C-OUT is also known as the majority 1's detector, whose output goes high when more than one input is high.
- A full adder logic is designed in such a manner that can take eight inputs together to create a byte-wide adder and cascade the carry bit from one adder to another.
- We use a full adder because when a carry-in bit is available, another 1-bit adder must be used since a 1-bit half-adder does not take a carry-in bit.
- A 1-bit full adder adds three 1 bit operands and generates 2-bit results.

Truth Table for full adder:

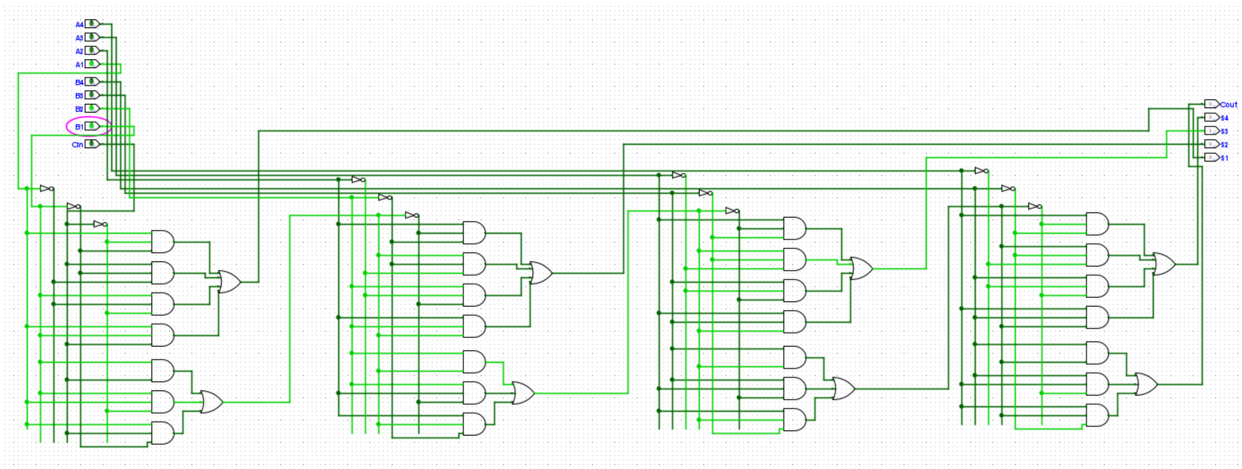
| Inputs | | | Outputs | |
|---|---|---|---|---|
| A | B | C – IN | Sum | C - Out |
| 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 1 | 0 |
| 0 | 1 | 0 | 1 | 0 |
| 0 | 1 | 1 | 0 | 1 |
| 1 | 0 | 0 | 1 | 0 |
| 1 | 0 | 1 | 0 | 1 |
| 1 | 1 | 0 | 0 | 1 |
| 1 | 1 | 1 | 1 | 1 |

A ripple carry adder is a digital circuit that adds two multi-bit binary numbers by cascading full adders in series, where the carry-out bit of each stage feeds directly into the carry-in of the next.

In this way from the least significant bit to the most significant bit the respective adders must wait for the previous adder to complete its calculations before doing its own calculation , which leads to a delay between input and output.

To create a 4 bit ripple carry adder we connect 4 full adders in a sequential manner

# Implementation:




# Results:

Successfully implemented a full adder , then a 4 bit ripple carry adder using only Basic gates